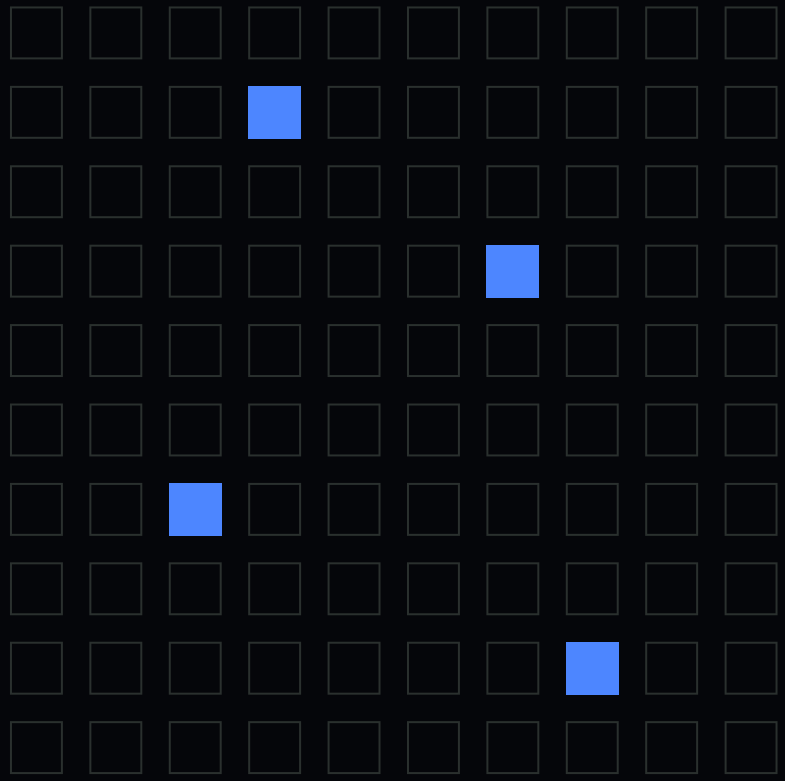
EIGHT CASES / ONE RULE / 11 NOVEMBER 2026

ONE AT A TIME

The world treats people and places in batches: it sprays the field, powers down the county, hands out the pills, closes the beach. Almost all the harm in that is collateral. Two machines show another way, and both are already working.



ONE HUNDRED UNITS. FOUR NEED THE HELP.

WHY THE WORLD IS BLUNT

Why the world is blunt

It is not cruelty. It is the cost of telling things apart. If you cannot see each case, you treat them all the same way, and the same way is always too much for some and too little for others.

IT CANNOT SEE EACH UNIT

You cannot check ten million people one at a time. So you check a sample, a queue, or whoever turns up.

IT CANNOT DECIDE PER CASE

A rule must be written before the case arrives. A rule that covers everyone has to fit the average, and nobody is average.

IT CANNOT ACT PER CASE

One switch, one sprayer, one prescription. The size of the tool sets the size of the harm.

THE METHOD

Two machines that already do it

THE LASER

A laser weeder sees every plant in the field and burns only the weed. Five thousand a minute, day or night, and the crop is never touched.

THE COW COUNTER

A camera in a barn doorway sees every cow, keeps hold of each one, and counts it once. In the dark. Even when they crowd together.

See every unit

A cheap camera reads each one, day or night.

Tell them apart

A model says: this one, not that one.

Act on the one

A small action, on that unit only.

Prove the rest

A record of what was left alone.

THE RULE

Look at every unit, cheaply. Decide one at a time. Act on that one. Then prove you left the rest alone.

THE GATE

Four conditions. Miss one and it fails.

This is not a tool you can point anywhere. It only works when all four are true, and the same test applies in a hospital, a war zone or a field.

01

The two things must differ

The weed and the crop have to be separable at the moment you act. If they look alike, you will hit the crop.

02

Looking must be cheap

Seeing every unit has to cost less than one miss. Otherwise you are back to sampling and guessing.

03

The action must stay small

A wrong act has to stop at the unit. If a false move takes out the street, you cannot aim.

04

The prize must be worth it

What you save has to be at least ten times the cost of deciding. Otherwise the maths does not close.

PART 1

The body

Four places where medicine treats a crowd, and the crowd is not the patient.

The X-ray reads first

NOW

1 in 5

of people who fell ill with TB in 2024 were never diagnosed. 1.23 million died that year.

WHO Global TB Report, 2025

ONE AT A TIME

430,000

chest X-rays read by software in seven countries in two years. They found 14,000 people with TB who had not coughed their way in.

Stop TB Partnership, 2024

HOW

A portable X-ray takes one image. Software scores it. Only the flagged go for the lab test.

PROVEN

WHO recommended computer-aided reading in 2021 and approved six products in June 2025, so the reading no longer needs a radiologist in the room.

WHAT IT CANNOT DO

Software scores a shadow, not a disease. Every flag still needs a lab test, and it is not approved for children under 15.

The eye test that travels

NOW

28.5m

people had the sight-threatening form of diabetic eye disease in 2020. 1.07 million were blind from it.

Ting et al., Ophthalmology, 2021; Vision Loss Expert Group, Eye, 2024

ONE AT A TIME

95%

of that blindness can be prevented if it is caught early. One photo at the diabetes visit grades the eye on the spot.

IAPB and IDF policy brief, 2024

HOW

A retinal camera takes one photo. Software grades it. Only the eyes needing a specialist are sent.

PROVEN

Thailand graded 7,651 patients at nine clinics with software: 91.4% sensitivity, above the specialists reading the same images, and the answer came back the same visit.

WHAT IT CANNOT DO

The camera needs a dark room, a trained photographer and a connection, and the disease only helps if laser or injections are reachable.

Name the snake first

NOW

1 vial

is licensed in India, a country with about 60 venomous snakes. It is raised against four of them.

Abraham et al., J. Emergencies, Trauma and Shock, 2025

ONE AT A TIME

95%

correct: Australia's bedside test names the snake in 15 to 25 minutes, so the right vial is opened, not the nearest one.

Johnston et al., Medical Journal of Australia, 2017

HOW

A swab from the bite, or urine, or blood. The test names the venom group, and the vial follows.

PROVEN

Ten years of national testing: the median dose fell from four vials to one, and deaths stayed flat at about two a year.

WHAT IT CANNOT DO

It types the venom after a doctor decides the bite is serious. It does not make that decision, and on mild bites 36% of tests read falsely positive.

The five-minute test

NOW

30%

of American antibiotic prescriptions, about 47 million a year, are unnecessary. Resistance killed 1.14 million people in 2021.

Fleming-Dutra et al., JAMA, 2016; GRAM Project, The Lancet, 2024

ONE AT A TIME

31%

of chest-infection visits got an antibiotic where doctors used a five-minute blood test, against 53% without it. Recovery was the same.

Cals et al., BMJ, 2009

HOW

A drop of blood, a reader, a reading in the room. A low score means no pill, and a plan to come back.

PROVEN

Twenty Dutch practices, 431 adults. Prescribing fell from 53% to 31%, and patient satisfaction did not move.

WHAT IT CANNOT DO

The reading rises with any inflammation. It trims pills for bronchitis but cannot pick the drug, and it needs a doctor willing to act on a low score.

PART 2

The warning

Two places where the alarm is loud, the area is vast, and the help cannot find the one who needs it.

Count who is still there

NOW

1 in 5

people in the tsunami zone did not evacuate, though the warning went out three minutes after the quake. 15,899 died.

Japan Cabinet Office survey, 2012; Japan National Police Agency, 2021

ONE AT A TIME

1 street

named for the search, because the count of who is still inside the zone says where to look.

Our design. No such count exists today.

HOW

Cameras at the roads in and out. The gap between them is who is left, street by street.

PROVEN

Zushi City ran eight cloud cameras over its beach in 2022 and 2023 and matched the officers' manual headcounts. Kamakura uses the same feeds after a tsunami.

WHAT IT CANNOT DO

A shore camera sees a few hundred metres. It cannot warn of a wave, only tell you who is still on the sand.

Trip one span

NOW

85

people died in the Camp Fire, started by power lines. The utility's answer was to cut power to whole counties for days.

CAL FIRE, 2025

ONE AT A TIME

1 switch

opens on the faulted span, and the hospital and the care home down the road stay on.

Utility fast-tripping practice, California, 2021 onward

HOW

Reclosers, cameras and satellites already exist. Two independent signals must agree.

PROVEN

Fast-tripping on high-fire spans is now standard in California, and the fault is located to the span, not the district.

WHAT IT CANNOT DO

In extreme fire weather the right answer is still to switch off. Precision lowers the cost of safety. It does not remove it.

PART 3

The ordinary

Two places where the blunt tool is so normal that nobody calls it harm: a pipe, a classroom.

Find the joint

NOW

346m m³

of water is lost every day, about $39 billion a year. The usual fix is to replace the street.

Liemberger and Wyatt, Water Supply, 2019

ONE AT A TIME

1 hole

instead of a street, because two microphones put a map point on the leak before the crew arrives.

Acoustic leak localisation, standard practice

HOW

Listeners clamped in valve chambers, two nights of sound on the same main.

PROVEN

Water utilities in the UK and Japan localise leaks this way today. The loggers cost a few hundred pounds each.

WHAT IT CANNOT DO

Two sensors cannot see a burst under a motorway. There you still need a camera and a person.

Test every child, not the class

NOW

70%

of 10-year-olds in low and middle income countries cannot read a simple story, and most never get tested.

World Bank, learning poverty, 2022

ONE AT A TIME

1 min

per child to find the level they actually read at, then teach in groups at that level, not by grade.

Teaching at the Right Level, Pratham

HOW

A tablet asks the child to read aloud. The score groups children by level, not by age.

PROVEN

Zambia grouped children by reading level in all 1,877 schools in two provinces. Children reading a paragraph rose from 68,892 to 110,228 in one year.

WHAT IT CANNOT DO

A test does not teach. Without a teacher free to regroup children, the score changes nothing.

THE LINE

Where this must never go

The ability that finds a weed also finds a person. A model that can tell one face from another can be pointed at a crowd, a border, a picket line. The line is simple: precision is for care, never for control.

POLICING

Predicting who might offend is the same maths pointed at a person. We do not build it.

WORK

A camera that scores a worker is surveillance with better manners. Precision belongs to the rota.

BORDERS

Counting people to push them back is not counting them to rescue them. The purpose decides.

WAR

The pattern is dual use by design, which is why the record of what it did not do has to be public.

Ranked by harm, then by proof

RANK	THE CASE	HARM PER YEAR	ALREADY PROVEN	TO PROVE IT
01	Antibiotic resistance	1.27m deaths a year	CRP trials, Europe	LOW
02	Tuberculosis	1.3m deaths a year	50+ countries	MEDIUM
03	Snakebite	81,000 to 138,000 deaths	Phone tools, trials	LOW
04	Learning	70% cannot read at 10	TaRL, at scale	LOW
05	Diabetic eye disease	1 in 3 diabetics	Thailand, national	MEDIUM
06	Evacuation	15,899 in one tsunami	Cameras, not yet joined	LOW
07	Water loss	$39bn a year	Standard practice	LOW
08	Wildfire grid	85 in one fire	California, in use	MEDIUM

START HERE

Three to prove first

01

Test every child

Cheapest to run, fastest to measure, and the result is a number a minister can repeat.

ONE TABLET PER SCHOOL

02

The five-minute test

The harm is already counted in prescriptions, so a fall in prescribing is the proof.

PER TEST

03

Count who is still there

Nobody has built it. One coastal town and one flood zone settle whether it works.

CAMERAS ALREADY INSTALLED

ONE LINE

Treat the case, not the category.

Every idea here starts with a batch that was cheaper than looking, and ends with a machine that looks anyway. The saving is not the machine. It is everything that was never hit.